

Figure 3.4 Our initial `buttonClick` function resizes the circles based on the numerical value of the associated attribute. The radius of each circle reflects the number of goals scored against each team, kept in the `ga` attribute of each datapoint.

```
teamG.on("mouseover", highlightRegion);
function highlightRegion(d) {
     d3.selectAll("g.overallG").select("circle")
            .style("fill", function(p) {
                  return p.region == d.region ? "red" : "gray";
            });
};
```

This time we used `d` as our variable, which is typical in the examples you'll see online for D3 functionality. As a result, we changed the inline function variable to `p`, so that it wouldn't conflict. Here you see an "ifsie," which is an inline `if` statement that compares the region of each element in the selection to the region of the element that you moused over, with results like those in figure 3.5.

Restoring the circles to their initial color on `mouseout` is simple enough that the function can be declared inline with the `.on` function:

```
teamG.on("mouseout", function() {
    d3.selectAll("g.overallG").select("circle").style("fill", "pink");
});
```

If you want to define custom event handling, you use `d3.dispatch`, which you'll see in action in chapter 9.

3.2.2 Graphical transitions

One of the challenges of highly interactive, graphics-rich web pages is to ensure that the experience of graphical change isn't jarring. The instantaneous change in size or color that we've implemented doesn't just look clumsy, it can actually prevent a reader from understanding the information we're trying to relay. To smooth things out a bit, I'll introduce transitions, which you saw briefly at the end of chapter 1.

Transitions are defined for a selection, and can be set to occur after a certain delay using `delay()` or to occur over a set period of time using `duration()`. We can easily implement a transition in our `buttonClick` function:

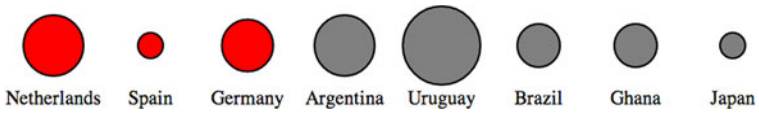


Figure 3.5 The effect of our initial `highlightRegion` selects elements with the same region attribute and colors them red, while coloring gray those that aren't in the same region.